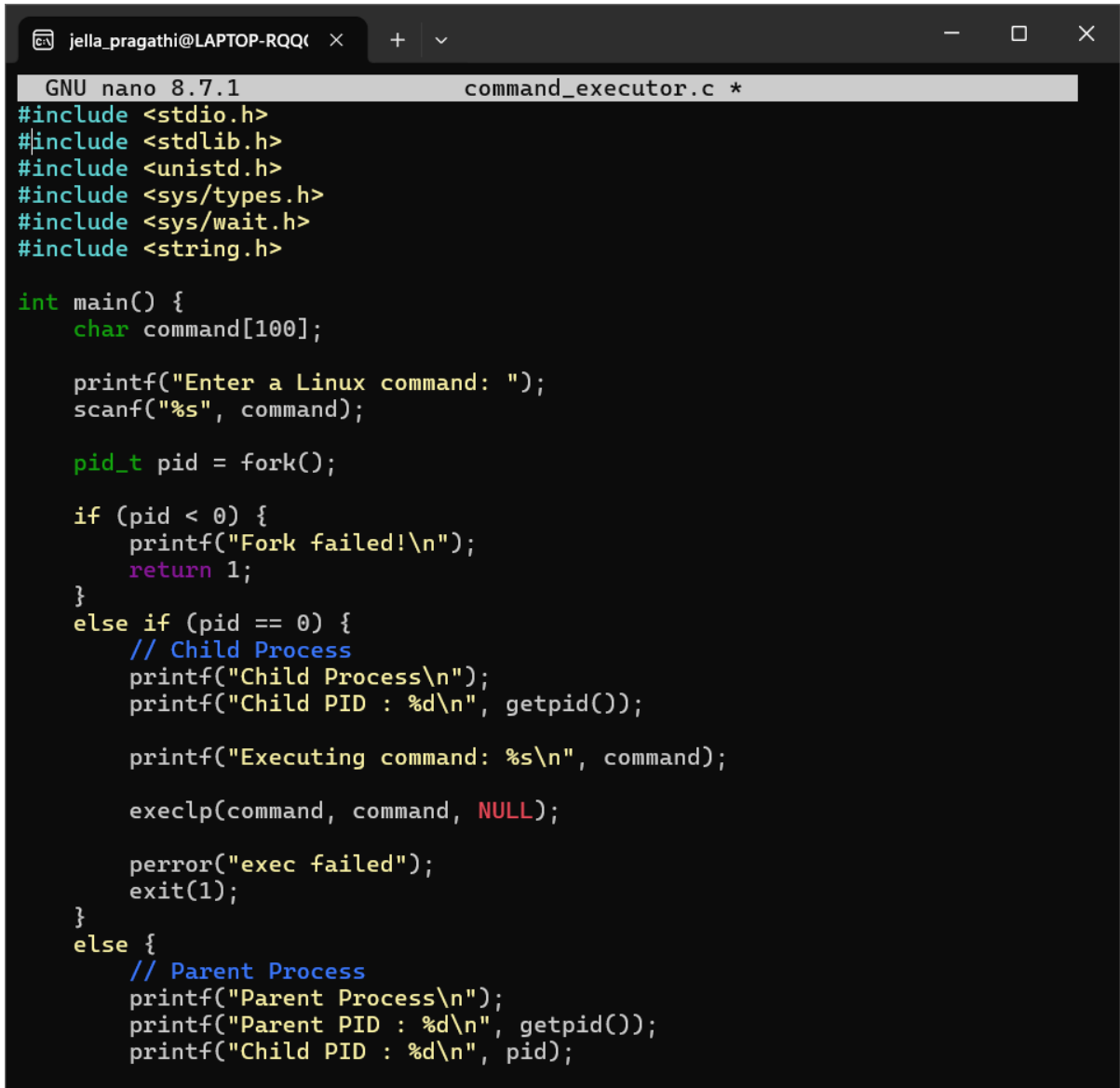


Practical week 2

Session 2:

Task 1:



```
GNU nano 8.7.1 command_executor.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }
    else if (pid == 0) {
        // Child Process
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());

        printf("Executing command: %s\n", command);

        execlp(command, command, NULL);

        perror("exec failed");
        exit(1);
    }
    else {
        // Parent Process
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
    }
}
```

```

        printf("Child process completed.\n");
    }

    return 0;
}

```

^{^G} Help ^{^O} Write Out ^{^F} Where Is ^{^K} Cut ^{^T} Execute
^{^X} Exit ^{^R} Read File ^{^_\} Replace ^{^U} Paste ^{^J} Justify

```

jella_pragathi@LAPTOP-RQQT555:~$ nano command_executor.c
jella_pragathi@LAPTOP-RQQT555:~$ gcc command_executor.c -o command_executor
jella_pragathi@LAPTOP-RQQT555:~$ ./command_executor
Enter a Linux command: ls
Parent Process
Child Process
Parent PID : 797
Child PID : 798
Child PID : 798
Executing command: ls
File1.txt      command_execution.c  parser      task2      task5
File2.txt      command_executor     parser.c    task2.c    task5.c
OSSP           command_executor.c  process     task3      tokenizer
buffer         history              process.c   task3.c    tokenizer.c
buffer.c       history.c            task1       task4
command_execution ossp_project      task1.c     task4.c
Child process completed.
jella_pragathi@LAPTOP-RQQT555:~$ |

```

Task 2:

```
jella_pragathi@LAPTOP-RQQG555:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               46 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      18
  On-line CPU(s) list:       0-17
Vendor ID:                   GenuineIntel
Model name:                  Intel(R) Core(TM) Ultra 5 125H
  CPU family:                6
  Model:                     170
  Thread(s) per core:        2
  Core(s) per socket:        9
  Socket(s):                 1
  Stepping:                  4
  Bogomips:                  5990.39
  Flags:                     fpvm de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt flu
                             sh_lld arch_capabilities

Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:         Microsoft
  Virtualization type:       full
Caches (sum of all):
  L1d:                       432 KiB (9 instances)
  L1i:                       576 KiB (9 instances)
  L2:                         18 MiB (9 instances)
  L3:                         18 MiB (1 instance)
NUMA:
```

```
jella_pragathi@LAPTOP-RQQG555:~$ cat /sys/devices/system/cpu/vulnerabilities/cpu_meltdown
jella_pragathi@LAPTOP-RQQG555:~$ cat /sys/devices/system/cpu/vulnerabilities/spectre_v1
jella_pragathi@LAPTOP-RQQG555:~$ cat /sys/devices/system/cpu/vulnerabilities/spectre_v2
jella_pragathi@LAPTOP-RQQG555:~$ cat /sys/devices/system/cpu/vulnerabilities/srbds
jella_pragathi@LAPTOP-RQQG555:~$ cat /sys/devices/system/cpu/vulnerabilities/tsa
jella_pragathi@LAPTOP-RQQG555:~$ cat /sys/devices/system/cpu/vulnerabilities/tsx_async_abort
jella_pragathi@LAPTOP-RQQG555:~$ cat /sys/devices/system/cpu/vulnerabilities/vmscape
jella_pragathi@LAPTOP-RQQG555:~$ ^C
jella_pragathi@LAPTOP-RQQG555:~$ uname -a
Linux LAPTOP-RQQG555 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
jella_pragathi@LAPTOP-RQQG555:~$ ps
  PID TTY          TIME CMD
 348 pts/0    00:00:00 bash
 855 pts/0    00:00:00 ps
```

```

855 pts/0    00:00:00 ps
jella_pragathi@LAPTOP-RQQGT555:~$ top
top - 08:04:55 up 12 min,  1 user,  load average: 0.02, 0.07, 0.02
Tasks: 25 total,  1 running, 24 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7651.2 total, 7029.5 free,  607.9 used,  165.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used. 7043.3 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   24200  15400 11560 S   0.0   0.2   0:00.80 systemd
    2 root        20   0    3180   2212  2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0    3180   2048  1940 S   0.0   0.0   0:00.00 init
   47 root        19  -1   50404  16864 15556 S   0.0   0.2   0:00.26 systemd-journal
   82 systemd+    20   0   22416  14464 11972 S   0.0   0.2   0:00.07 systemd-resolve
   90 root        20   0   35452  12328  9148 S   0.0   0.2   0:00.56 systemd-udev
  108 root        20   0    2888   1948  1820 S   0.0   0.0   0:00.05 chronyd-starter
  109 root        20   0    4472   3152  2904 S   0.0   0.0   0:00.00 cron
  110 message+    20   0    8796   5392  4660 S   0.0   0.1   0:00.09 dbus-daemon
  127 root        20   0   44096  29732 14592 S   0.0   0.4   0:00.29 networkd-dispat
  166 root        20   0   18440   9408  8160 S   0.0   0.1   0:00.10 systemd-logind
  196 syslog      20   0   220548  5928  4616 S   0.0   0.1   0:00.08 rsyslogd
  224 _chrony      20   0   23840  11324  7172 S   0.0   0.1   0:00.14 chronyd
  240 _chrony      20   0   12096   2408  1760 S   0.0   0.0   0:00.02 chronyd
  257 root        20   0    5492   2720  2536 S   0.0   0.0   0:00.00 agetty
  260 root        20   0  123460  32496 17888 S   0.0   0.4   0:00.20 unattended-upgr
  346 root        20   0    3184   1116   980 S   0.0   0.0   0:00.00 SessionLeader
  347 root        20   0    3200   1252  1108 S   0.0   0.0   0:00.07 Relay(348)
  348 jella_p+    20   0    6268   5608  3928 S   0.0   0.1   0:00.09 bash
  349 root        20   0    8648   5572  4728 S   0.0   0.1   0:00.02 login
  393 jella_p+    20   0   22344  13572 10956 S   0.0   0.2   0:00.18 systemd
  395 jella_p+    20   0   22924   4096  2092 S   0.0   0.1   0:00.00 (sd-pam)
  423 jella_p+    20   0    6136   5424  3864 S   0.0   0.1   0:00.03 bash
  624 polkitd     20   0  306444   8428  7616 S   0.0   0.1   0:00.09 polkitd
  866 jella_p+    20   0    8172   6056  3936 R   0.0   0.1   0:00.12 top

```